Basic Information

Experiment No.: 06

Date: 11-10-2026

Title: Longest Substring Without Repeating Characters

Objective / Aim

To write a Java program to find the length and the longest substring without repeating characters in a given string.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

- Java JDK
- Any Java IDE or Command Prompt
- Operating System: Windows/Linux

Algorithm / Logic Description

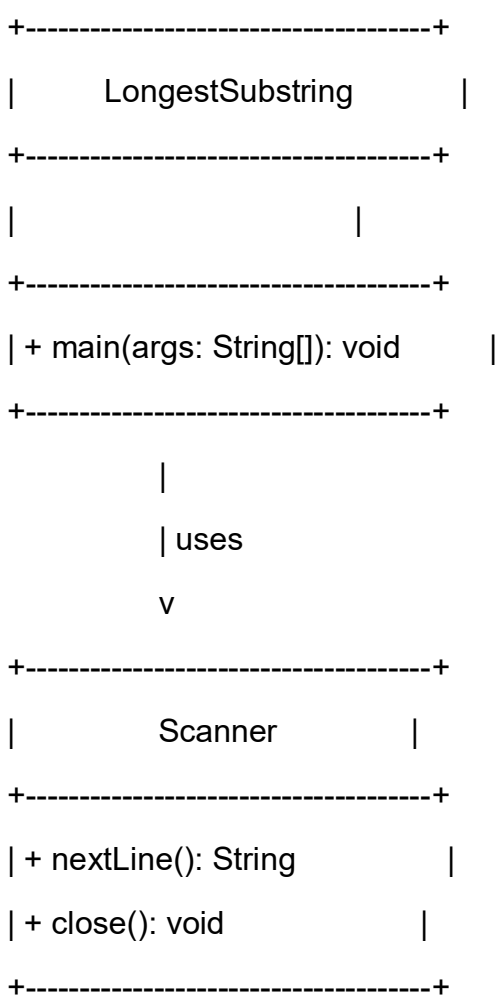
1. Read a string from the user.
2. Initialize the starting position and maximum length to zero.
3. Traverse each character of the string.
4. Check whether the current character is repeated in the current substring.
5. If repeated, move the starting position forward.

6. Update the maximum length and substring when a longer substring is found.

7. Display the length and substring.

8. End the program.

📐 UML Class Diagram



Variables inside main():

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s : String - Input string

max : int - Maximum substring length

start : int - Current substring start

begin : int - Best substring start

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i, j : int - Loop counters

result : String - Longest substring

💻 Source Code

```
import java.util.Scanner;

public class LongestSubstring {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Read the input string
        System.out.print("Enter a string: ");
        String s = sc.nextLine();

        int start = 0, max = 0, begin = 0;

        // Check each character in the string
        for (int i = 0; i < s.length(); i++) {

            // Check for repeated characters
            for (int j = start; j < i; j++) {
                if (s.charAt(j) == s.charAt(i)) {
                    // Move start after the repeated character
                    start = j + 1;
                    break;
                }
            }

            // Update the longest substring details
```

```
            if (i - start + 1 > max) {
                max = i - start + 1;
                begin = start;
            }
        }

        // Extract the longest substring
        String result = s.substring(begin, begin + max);

        // Display the result
        System.out.println("Length: " + max);
        System.out.println("Substring: " + result);

        sc.close();
    }
}
```

🔨 Compilation & Execution Commands

Save the file as:

LongestSubstring.java

Compile:

javac LongestSubstring.java

Run:

java LongestSubstring

📟 Output Screen

Example 1

Enter a string: abcabcbb

Length: 3

Substring: abc

Example 2

Enter a string: pwwkew

Length: 3

Substring: wke

📌 Result:

The Java program was successfully implemented to find the length and longest substring without repeating characters in a given string.